

Business and model summary

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Total transactions: 20,000
Fraud transactions: 339
Fraud rate: 0.0169 (1.70%)
Total transaction volume: 2,648,491.94
Average transaction amount: 132.42
Total fraud volume: 61,443.70
Average fraud transaction amount: 181.25
Average non-fraud transaction amount: 131.58
Fraud volume share: 0.0232 (2.32%)
Average velocity (fraud): 28.97
Average velocity (non-fraud): 19.65
Average merchant risk (fraud): 51.39
Average merchant risk (non-fraud): 37.16
Fraud rate for new merchants: 0.0350
Fraud rate for existing merchants: 0.0116
Fraud rate for foreign transactions: 0.0589
Fraud rate for domestic transactions: 0.0142

Top card types by fraud rate:

Amex: 0.0201
Discover: 0.0196
Visa: 0.0178
Mastercard: 0.0154
RuPay: 0.0134

Top merchant categories by fraud rate:

Crypto Exchange: 0.0499
Gift Cards: 0.0438

Gaming: 0.0367
Electronics: 0.0236
Streaming: 0.0208

Model evaluation

Accuracy: 0.9830
ROC AUC: 0.8967
Average precision: 0.2344

Classification report:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	19661
1	0.00	0.00	0.00	339
accuracy			0.98	20000
macro avg	0.49	0.50	0.50	20000
weighted avg	0.97	0.98	0.97	20000

Confusion matrix:

[[19661, 0],
[339, 0]]

Top feature importances:

feature	importance
merchant_risk_score	0.063336
velocity_score	0.060805
account_balance_usd	0.045385

amount_usd	0.045370
amount_log	0.044916
distance_from_home_km	0.044237
cvv_retry_count	0.043746
balance_ratio	0.043720
customer_age	0.037478
card_age_months	0.037238